

Angel Verde Salas

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EDUCATION

Texas State University

Expected graduation: Summer 2026 , San Marcos, TX

B.S. Mathematics,
B.S. Applied Mathematics,
BS. Computer Science

Master's in Mathematics

Starting, Fall 2026

SKILLS

- | | |
|---------------------------------------|-------------------------|
| ▪ Python; NumPY, Pandas | ▪ Linux, Git, Jira |
| ▪ HTML, CSS, JavaScript; React Native | ▪ MATLAB, R |
| ▪ Mathematical Thinking | ▪ Fluent in Spanish |
| ▪ Excel, Word, PowerPoint | ▪ C++ |
| ▪ Statistics & Data Science | ▪ Mathematical modeling |

WORK EXPERIENCE

NSF Math Researcher: Statistics and Mathematical Modeling

Summer 2024

Prairie View A&M University NSF REU: Mathematical Modeling in the Sciences

Prairie View, TX

- As an NSF undergraduate researcher working with Dr. Wickramasinghe, I investigated the efficiency of an approximation method for various probability distributions, including Normal, Exponential, Poisson, and Binomial. I derived mathematical formulas for both single-variable and multivariate cases and visualized the functions using R. My analysis revealed that the approximation technique achieved high accuracy, ranging from the upper 90s to 100%.

Undergraduate Instructional Assistant – Pre-Calculus, Calculus I, Calculus II

Aug. 2023 –Present

Texas State Math Department

San Marcos, TX

- In my role as an Undergraduate Instructional Assistant, I supported classroom activities, offered individualized assistance, managed grading, developed supplementary materials, and conducted office hours. This experience improved my communication skills, leadership abilities, and deepened my mathematical thinking, all while contributing to a productive learning environment for students.

Math Cats Tutor, Learning/Class Assistant

Jan. 2023– Present

Texas State Math Department

San Marcos, TX

- As a Math CATS tutor, I provide valuable assistance to fellow students across a spectrum of mathematical subjects. My role not only supports their learning but also enriches my own comprehension of mathematics. As a Lab\Class Assistant my role revolves around providing valuable support and ensuring students grasp the core concepts of the course. This dual benefit underscores my commitment to both personal growth and academic success

PROJECTS

Fitness Website

- Worked on a group project creating a fitness website called [HealthFitnessPro](#), where I led the UX/UI design and frontend implementation for the nutrition, account, and workout history modules using React. I architected the underlying database schemas and data models for the custom workout creator and exercise library, ensuring seamless integration with the FastAPI backend. Additionally, I developed a modular frontend architecture with reusable components and established comprehensive unit testing plans to ensure application stability.

Predicting Player Performance from FIFA Attributes in R

- Worked on a group project analyzing FIFA player attribute data to predict the overall ratings of youth soccer players. Used PCA to reduce dimensionality and visualize patterns in player traits, followed by Lasso regression, stepwise selection, and decision trees to identify key predictors by player position. Lasso consistently outperformed other models, and the project highlighted valuable traits for scouting and talent evaluation

Image Compression Using FFT, DCT, and DWT in Python and MATLAB

- Explored frequency-domain image compression by applying Fast Fourier Transform (FFT), Discrete Cosine Transform (DCT) and Discrete Wavelet Transform (DWT) to grayscale images using Python and MATLAB. Implemented low-pass filtering to retain low-frequency components and reconstructed images via inverse transforms. Finally comparing their respective compressed images using file size, visual quality, and runtime.

Secure Cancer Prediction System with Homomorphic Encryption

- Extended an open-source [MLE framework](#) by implementing and empirically comparing BFV and CKKS homomorphic encryption schemes (Microsoft SEAL/C#) for privacy-preserving cancer prediction. Evaluated accuracy and runtime trade-offs using encrypted genomic datasets (~5,600 features, 22 cancer types), highlighting differences between exact and approximate encrypted inference.

LWE Parameter Benchmarking & Error Estimation

- Currently analyzing the [Learning With Errors \(LWE\)](#) problem by simulating how variations in error distributions and vector dimensions affect encryption integrity. Using the [TAPAS dataset](#), mapping the trade-off between decryption success and cryptographic hardness to identify the optimal noise levels and vector dimensions required for secure two-party communication.

LEADERSHIP & INVOLVEMENT

Secretary, Society for Industrial and Applied Mathematics

08/25-Present, Texas State University

- Maintained meeting minutes, managed internal communications, coordinated events, and organized documentation for club activities.

Vice president, Technical Interview Prep

08/25-Present, Texas State University

- Organized and led technical interview prep sessions focused on algorithms, data structures, and coding practice.